

Economic Dispatch Incorporating Wind Energy by BH, BBO and DE

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Abstract—This paper proposes a group of Evolutionary Algorithms: Black Hole (BH) algorithm, Biogeography Based Optimization (BBO) algorithm and Differential Evolution (DE) algorithms to solve the economic dispatch problem incorporating the wind energy (EDIW) uncertainty. The proposed methods have a simple procedure to find a near optimal solution for the non-smooth and non-convex problems. There are many constraints in the economic dispatch problem such as generators capacity, ramp rate limit, and prohibited operation zones. There is also the problem of valve point effect which will change the fuel cost equation of the generators, which categorize it as non-smooth and non-convex problem. In this paper, BH, BBO and DE are employed to solve the EDIW problem for three and six generator system. Results of the three algorithms will be compared with other used algorithms.

Index Terms—Evolutionary Algorithms, Economic Dispatch, Wind Energy .

I. INTRODUCTION

Economic dispatch (ED) is finding the minimum cost of generating electrical energy while satisfying the load demand and the system constraints. There are many constraints in economic dispatch problem such as generators capacity, ramp rate limit, prohibited operation zone [1]. Modern generation unit has input-output characteristics which are highly nonlinear because of the valve point effect and ramp rate limits and discontinuous due to prohibited operation zones [2]. Classical proposed solutions such as lambda iteration method, gradient method, base point, and participation factor method as mentioned in [3] assume that the cost curve is continuous and monotonically increasing (convex and smooth objective). This assumption may lead to unfeasible solutions or may lead to suboptimal solution which means that the conventional methods fails in solving such problems with non-smooth and non-convex objectives. Heuristic optimization algorithm such as Genetic Algorithm [4], Particle Swarm Optimization algorithm [5], Biogeography Based Optimization algorithm [1], [6], etc. propose a feasible solution to the problem because they impose no restriction on the nature of the objective functions or constraints. They also provide fast, easy and optimal or near optimal solution. Injection of wind Power in the power network has advantageous such as reducing of fuel cost and reducing the emissions [7] and wastes which destroy the environment. On the other hand it introduces challenges in the suitable way to model the wind turbine output intermittency nature because it depends on the wind speed. Many algorithms proposed in literature to deal with intermittent behavior of the wind turbines. Conventional approach models the intermittent resources as negative load [8]. Model Predictive Control

proposed in [9] is used for dispatching the available generation with the objective of minimizing the total production cost. A probabilistic model for the wind turbine output based on the Weibull probability density function is used in [10], which account for both overestimation and underestimation of available wind power. [11] use the probabilistic model and solves economic load dispatch problem with random wind power constraint using penalty function and interior point methods. This paper presents the application of group of Evolutionary algorithm to solve the problem of EDIW with generators capacity, ramp rate limit, and prohibited operation zones as constraints. Results of each algorithm will be compared with other used algorithms. This paper is organized as follows. In Section (II), mathematical formulation of the EDIW problem are reviewed. In Section (III), the Evolutionary algorithm is described, while Section (IV) present the numerical results. Finally, Section (V) presents the conclusion.

II. MATHEMATICAL FORMULATION

The general formulation of the EDIW optimization problem can be stated as follows [12]:

$$\min FC + WC \quad (1)$$

where

$$FC = \sum_{i=1}^{N_x} a_i P_i^2 + b_i P_i + c_i \quad (2)$$

Where FC is the total fuel cost of generators, N_x is the number of generator units, P_i is the output power of generation unit i , a_i , b_i and c_i are the fuel cost coefficients. The fuel cost equation considering the valve point effect is given by:

$$FC = \sum_{i=1}^{N_x} a_i P_i^2 + b_i P_i + c_i + |e_i \sin(f_i (P_i^{min} - P_i))| \quad (3)$$

Where e_i , f_i are the fuel cost coefficients and P_i^{min} is the minimum active power output of the i^{th} generating unit. The total wind cost is compose of the sum of three components: direct cost, underestimate cost and overestimate cost as shown in (4).

$$WC = \sum_{i=1}^{N_w} C_{di} + C_{pi} + C_{ri} \quad (4)$$

where WC is the total cost of wind power, N_w is the number of wind turbines, C_{di} is direct cost for wind turbine

i , C_{pi} is underestimate cost for wind turbine i and C_{di} is overestimate cost for wind turbine i . A detail total wind power cost is shown in (5), which show that there are two components that depend on the expected value of exceeding or not exceeding the scheduled value. In both cases there will be cost to be paid, hence there will be an effect on the total cost.

$$WC = \sum_{i=1}^{Nw} d_i w_i^{sc} + kp_i E_i^{ue} + kr_i E_i^{oe} \quad (5)$$

where w_i^{sc} is the scheduled wind power for the i^{th} wind turbine, w_i available wind power for the i^{th} wind turbine, E_i^{ue} is the expected value of $w_i > w_i^{sc}$ for the i^{th} wind turbine, E_i^{oe} is the expected value of $w_i < w_i^{sc}$ for the i^{th} wind turbine, d_i , kp_i and kr_i are the available, penalty, and reserve cost coefficients for the i^{th} wind turbine respectively.

$$E_i^{ue} = \int_{w_i^{sc}}^{w_r} (w - w_i^{sc}) f_w(w) dw \quad (6)$$

$$E_i^{oe} = \int_0^{w_i^{sc}} (w_i^{sc} - w) f_w(w) dw \quad (7)$$

where $f_w(w)$ is the transferred Weibull distribution from wind speeds to wind power. The cost model used in this paper is will explained in [13].

A. Equality constraints

1) *Real Power Balance Constraint*: Equation (8) represent the equality constraint in the problem, where the P_D represent the total load demand and P_L is the power losses.

$$\sum_{i=1}^{Nw} w_i + \sum_{i=1}^{Nx} P_i = P_D + P_L \quad (8)$$

$$P_L = \sum_{i=1}^N \sum_{j=1}^N P_i B_{ij} P_j + B_{0i} P_i + B_{00} \quad (9)$$

Where B_{ij} , B_{0i} and B_{00} are the Power loss coefficients.

B. Inequality constraints

1) *Generator Power Limit Constraint*: Generation units are restricted by their limits in term of generation active power as follows:

$$P_i^{min} \leq P_i \leq P_i^{max} \quad i = 1, 2, \dots, Nx \quad (10)$$

where P_i^{min} , P_i^{max} , are the minimum active power output, maximum active power output of the i^{th} generating unit respectively.

2) *Ramp Rate Limit Constraint*:

$$\begin{aligned} P_{i \text{ now}} - P_{i \text{ new}} &\leq DR_i \\ P_{i \text{ new}} - P_{i \text{ now}} &\leq UR_i \end{aligned} \quad (11)$$

where DR_i is the down ramp limit of the i^{th} generator and UR_i is the up ramp limit of the i^{th} generator.

3) Prohibited Operating Zone Constraint:

$$\begin{aligned} P_i^{min} &\leq P_i \leq P_{i,1}^l \\ P_{i,1}^u &\leq P_i \leq P_{i,3}^l \\ P_{i,3}^u &\leq P_i \leq P_i^{max} \\ i &= 1, 2, \dots, Nx \\ j &= 1, 2, \dots, m_i \end{aligned} \quad (12)$$

where $P_{i,j}^l$ is the lower and upper bound of the j^{th} prohibited zone of the i^{th} generation unit respectively, and m_i is the total number of prohibited operating zones of the i^{th} generation unit.

4) Wind turbine Power Limit Constraint:

$$0 \leq w_i \leq w_i^r \quad i = 1, 2, \dots, Nw \quad (13)$$

where w_i^r , is the rated active power output of the i^{th} wind turbine unit.

III. EVOLUTIONARY ALGORITHMS

A. black hole algorithm

The black hole algorithm (BH) is a population based algorithm inspired by the black hole phenomenon [14] and has common features with other population based algorithms such as genetic algorithm(GA), particle swarm optimization(PSO), differential evolution (DE), etc. A candidate solution in the BH algorithm is represented by a star and the best candidate solution is selected to be as black hole. Two operations in the proposed algorithm are used to evolve the population toward the optimality: star absorption and star sucking.

1) *Star absorption*: It is the operation of moving the stars towards the black hole under the black hole absorbing force and in the optimization method it is equivalent to recombination in other evolutionary algorithms. For example crossover in GA, crossover in DE, migration in BBO, etc. Star absorption can be formulated as follows:

$$x_i^{t+1} = x_i^t + r (x_{BH} - x_i^t) \quad (14)$$

where x_i^{t+1} and x_i^t are the location i^{th} star (candidate solution) at iteration t and $t + 1$ respectively, X_{BH} is the location of the black hole(best candidate solution) and r is random number in the interval $[0,1]$.

2) *Star sucking*: During the moving operation of stars toward the black hole, there is a probability of crossing the event horizon which is the point of no return. If a star (candidate solution) crosses the event horizon it will be sucked by the black hole (best solution) and the new solution will be generated in order to keep a constant number of candidate solutions. The radius of the event horizon in the BH algorithm is calculated as follows:

$$R = \frac{f_{BH}}{\sum_{i=1}^N f_i} \quad (15)$$

where f_{BH} is the fitness value of the black hole, f_i is the fitness value of the i^{th} star and N is the number of stars. If the distance between the candidate solution and the best solution

is less than R , that candidate solution will be replaced by new solution. The pseudo code for the BH algorithm can be found in [15] .

B. Biogeography Based Optimization

Biogeography based optimization (BBO) method is a population based algorithm proposed by Simon in [16]. BBO is inspired by the study of distribution of species in Habitats. The mathematical model was developed by MacArthur and Edward Wilson as pointed in [16]. In BBO Islands represent a feasible candidate solution. The solution will be ranked using two variables: immigration rate λ (bad) and emigration μ (good) rate. The fitness is defined in BBO as the habitat suitability index (HSI) and there are two main operations in this algorithm: migration and mutation. The pseudo code for the BH algorithm is shown in Algorithm (1).

1) *Migration*: Migration is the process of probabilistically sharing information between habitat using the λ and μ rates. We modify each solution based on other solutions by first randomly selecting an island to be modified (poor island) using the immigration rate λ (bad). After that using the emigration rate μ (good) we select a source of modification (good island). Then we randomly select a feature from the good island to modify the poor island.

2) *Mutation*: a rarely severely destructive events can extremely change the HSI of a habitat and can cause the species count to deviate from its equilibrium point [16]. In mutation a randomly new candidate solution will be generated to replace the old one using the selected mutation rate.

Algorithm 1 BBO Algorithm

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1: Initialize the BBO parameters
2: Generate a random set of habitats(  $I_i$  and  $i \in [1, N]$ )
3: Calculate the fitness (HSI) of each habitat
4: Calculate and map  $\mu$  and  $\lambda$ 
5:  $I_{Best} \leftarrow$  Optimal solution
6: while not (termination criterion) do
7:   Migration process
8:   Mutation process
9:   Fitness,  $\mu$  and  $\lambda$  Calculation and mapping
10:   $I_{Best} \leftarrow$  Optimal solution
11: end while
12: return

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C. Differential Evolution

DE is a population based optimization algorithm introduced by Ken Price and Rainer Storn [17]. The motivation of the algorithm was a result of developing a solution to the Chebyshev polynomial coefficients and the optimization of digital filter coefficients. This algorithm uses three operations to generate new candidate solutions: mutation, crossover and selection [18].

TABLE I
PARAMETERS FOR OPTIMIZATION ALGORITHMS

Test system	1			2		
Algorithm	co	pf	F	co	pf	F
BBO	-	0.005	-	-	0.005	-
DE	0.5	0.005	0.9	0.5	0.005	0.9

TABLE II
WIND TURBINE DATA

k	c	k_p	k_r	d	V_i	V_r	V_0	W_r
2	10	6	10	8	4	12.5	20	165

1) *Mutation*: In the mutation process a new candidate solution is created by modifying a randomly selected solution as shown generally in the Equation (16):

$$v_i = \begin{cases} x_{r_1} + F \times (x_{r_2} - x_{r_3}) & r < pf \\ x_{r_1} + \frac{(F+1)}{2} \times (x_{r_2} - x_{r_1} + x_{r_3} - x_{r_1}) & r \geq pf \end{cases} \quad (16)$$

Where F is the step size parameter, pf is the mutation rate, r_i is random integer and x_{r_i} is the candidate solution in the population which has a position r_i .

2) *Crossover*: In the crossover process the old and new solution generated in mutation are merged as shown in (17) to generate new population.

$$u_i = \begin{cases} v_i & r < co \\ x_i & r \geq co \end{cases} \quad (17)$$

Where co is the crossover probability.

3) *Selection*: The selection process is done by implementing (18) which choose between the parents and the children based on the fitness function.

$$x_i = \begin{cases} u_i & f(u_i) < f(x_i) \\ x_i & f(u_i) \geq f(x_i) \end{cases} \quad (18)$$

IV. SIMULATION RESULTS

For all the test cases the software has been written in MATLAB 8.2 environment and executed on 2.9 GHz Intel core I7-3520M CPU with 8 GB RAM. For all the algorithms, the selected number of generation, population size and number of iteration for the first system systems are 100, 100 and 20 respectively. On the other hand the selected number of generation, population size and number of iteration for the second system systems are 100, 200 and 20 respectively.

A. Test system 1

This system is composed of three generation units with quadratic cost functions and the effects of valve-point loading is considered. All the equality and inequality constraints are considered except the ramp rate limit and the prohibited operation zones. The unit characteristics of cost coefficients, and generators operating limits were taken from [19]. The evolutionary algorithm parameters and the wind turbine data

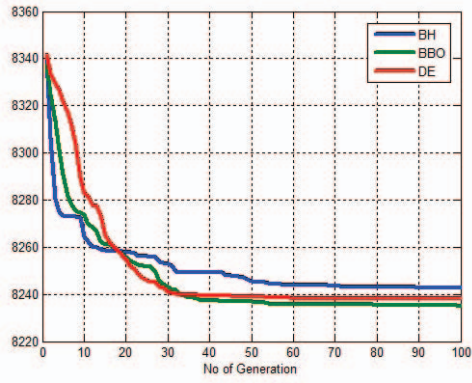


Fig. 1. Mean Convergence Characteristics of minimum cost obtained by using BH, BBO and DE for test system 1 (WPF=0).

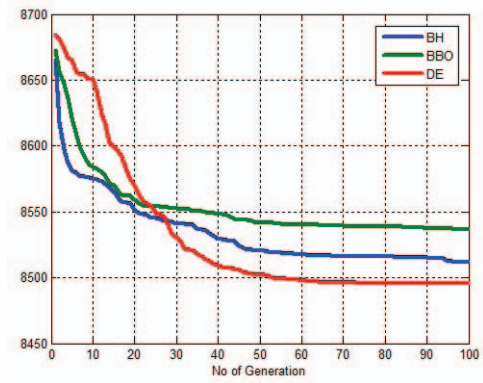


Fig. 3. Mean Convergence Characteristics of minimum cost obtained by using BH, BBO and DE for test system 1 (WPF=165).

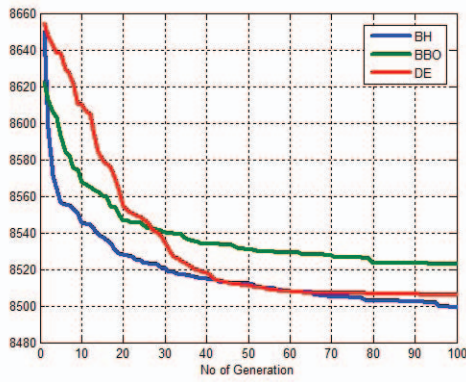


Fig. 2. Mean Convergence Characteristics of minimum cost obtained by using BH, BBO and DE for test system 1 (WPF=50).

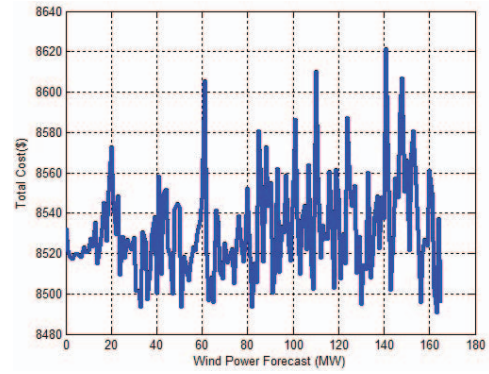


Fig. 4. Total cost variation with respect to wind power generation for test system 1.

is shown in Tables (I) and (II) respectively. The result obtained using the BH, BBO and DE are shown in Tables (III) and (IV) for three different wind power forecasts. Table (III) shows that the three algorithms succeed in obtaining a feasible solution which is close to each other. As shown in Table (IV), in the first forecast point (0MW), BH is higher than other by 0.08% in minimum cost, and in average cost BBO is better than BH and DE by 0.09% and by 0.04% respectively. While, in the second point (50MW) BH and DE are better than BBO algorithm in minimum cost by 0.05% and in average cost BH is lower than BBO and DE by 0.28% and 0.08% respectively. Finally, in the last forecast point (165MW), DE proposes a best cost compared to other algorithm in all the cases. The convergence characteristic of the mean cost is shown in Fig.1 to Fig.3 for the three different wind power forecast. Fig.4 shows the total cost variation considering the valve point effect as a function of wind forecasts based on the wind turbine power probabilistic model. This figure clearly shows that the minimum cost will not be necessary obtained as the wind generation increase and this this figure result from the effect of the valve point effect and the wind power cost model.

B. Test system 2

This test system consists of six generation units with quadratic cost functions, transmission losses, ramp rates limits and prohibited operating zones of these thermal units. The unit characteristics of cost coefficients, and generators operating limits were taken from [5]. The evolutionary algorithm parameters and the wind turbine data is shown in Table (I) and Table (II) respectively. The result obtained using the proposed algorithms are shown in Table (V) and Table (VI). In the first forecast point (0MW), all the algorithm gets the same result in minimum and average cost as shown in Table (VI). DE provide a slightly better minimum and average cost than other algorithm in the second point. In the third forecast point, BBO is higher than both algorithm by 0.004% in minimum cost and in average cost BH is better than BBO and DE by 0.01% and 0.004% respectively. Fig. 5 to Fig. 6 show the convergence curve of the mean cost for the three different wind power forecast. Fig. 8 shows the variation of the total cost with the wind energy forecast. It is obvious that the minimum cost will not be necessary obtained as the wind generation increase. This is figure result from the effect of the prohibited operation zones, ramp rate limit and the wind power cost model.

TABLE III
OPTIMAL RESULTS FOR TEST SYSTEM 1 ($P_D = 850MW$)

Wind Power Forecast	0 MW			50 MW			165 MW		
Algorithm	BH	BBO	DE	BH	BBO	DE	BH	BBO	DE
P_1 (MW)	498.93	300.25	300.27	399.22	399.55	399.20	399.20	399.06	399.20
P_2 (MW)	99.87	149.75	149.73	99.87	100.00	99.87	99.88	99.84	99.87
P_3 (MW)	251.20	399.99	400.00	324.41	324.38	324.40	324.40	324.08	324.40
P_W (MW)	0.00	0.00	0.00	26.51	26.07	26.53	26.51	27.02	26.53
Fuel Cost (\$/h)	8241.18	8234.08	8234.07	7954.80	7963.18	7954.28	7954.69	7954.11	7954.28
C_d (\$/h)	0.00	0.00	0.00	212.06	208.56	212.28	212.10	216.12	212.28
C_p (\$/h)	296.35	296.35	296.35	206.51	207.75	206.43	206.50	205.09	206.43
C_r (\$/h)	0.00	0.00	0.00	115.33	113.01	115.48	115.36	118.04	115.48
Total cost (\$/h)	8537.53	8530.44	8530.43	8488.70	8492.50	8488.47	8488.64	8493.37	8488.47

TABLE IV
COMPARISON OF ALGORITHMS RESULTS OF TEST SYSTEM 1

Wind Power Forecast	0MW			50MW			165MW		
Algorithm	BH	BBO	DE	BH	BBO	DE	BH	BBO	DE
Minimum total cost (\$/h)	8537.53	8530.44	8530.43	8488.70	8492.50	8488.47	8488.64	8493.37	8488.47
Average total cost (\$/h)	8538.94	8531.44	8534.51	8499.38	8523.05	8506.03	8511.85	8536.74	8495.58
Maximum total cost (\$/h)	8564.65	8548.01	8541.63	8577.35	8564.27	8566.12	8570.47	8585.04	8531.14
Standard deviation (\$/h)	6.05	3.90	3.83	25.29	20.60	24.68	34.71	26.11	14.79

TABLE V
OPTIMAL RESULTS FOR TEST SYSTEM 2 ($P_D = 1263MW$)

Wind Power Forecast	0 MW			50 MW			165 MW		
Algorithm	BH	BBO	DE	BH	BBO	DE	BH	BBO	DE
P_1 (MW)	447.64	447.57	447.39	438.77	435.58	437.00	435.98	436.97	436.06
P_2 (MW)	173.35	173.58	173.22	166.75	164.49	165.90	164.68	165.55	164.44
P_3 (MW)	263.48	263.52	263.41	255.79	256.96	255.97	254.45	259.37	254.34
P_4 (MW)	138.56	138.00	139.00	129.59	126.76	130.56	129.44	127.05	129.81
P_5 (MW)	165.35	165.79	165.39	156.84	153.00	156.95	156.36	156.07	156.41
P_6 (MW)	87.08	87.00	87.04	79.38	88.00	78.16	77.92	73.00	77.85
P_W (MW)	0.00	0.00	0.00	47.47	49.75	49.99	55.61	56.49	55.52
P_L (MW)	12.45	12.47	12.44	11.60	11.55	11.54	11.44	11.50	11.44
Fuel Cost (\$/h)	15443.08	15443.09	15443.08	14803.86	14774.43	14770.06	14694.93	14683.75	14696.09
C_d (\$/h)	0.00	0.00	0.00	379.78	398.04	399.96	444.89	451.90	444.19
C_p (\$/h)	296.35	296.35	296.35	156.40	151.93	151.47	141.23	139.72	141.38
C_r (\$/h)	0.00	0.00	0.00	241.47	256.84	258.47	297.56	303.81	296.95
Total cost (\$/h)	15739.43	15739.44	15739.43	15581.51	15581.24	15579.96	15578.61	15579.18	15578.61

TABLE VI
COMPARISON OF ALGORITHMS RESULTS OF TEST SYSTEM 2

Wind Power Forecast	0MW			50MW			165MW		
Algorithm	BH	BBO	DE	BH	BBO	DE	BH	BBO	DE
Minimum total cost (\$/h)	15739.43	15739.44	15739.43	15581.51	15581.24	15579.96	15578.61	15579.18	15578.61
Average total cost (\$/h)	15739.52	15739.52	15739.56	15583.90	15583.00	15580.96	15578.74	15580.30	15579.41
Maximum total cost (\$/h)	15739.80	15739.86	15741.75	15586.51	15590.81	15583.95	15579.03	15582.04	15583.10
Standard deviation (\$/h)	0.09	0.10	0.52	1.23	2.06	1.23	0.12	0.83	1.23

V. CONCLUSION

In this paper, black hole (BH) algorithm, biogeography based optimization algorithm (BBO) and differential evolution (DE) are proposed and successfully applied to solve the economic dispatch problem incorporating the wind energy (EDIW) uncertainty. The algorithms were applied on two systems, one is composed of three generation units with quadratic cost functions and the effects of valve-point loading is considered and the second consists of six generation units with quadratic cost functions, transmission losses, ramp rates limits and prohibited operating zones. The simulation results

of the proposed algorithms have been compared with each other. Based on results obtained in both cases, the minimum cost will not be guaranteed as the wind power forecast simply increase based on the wind power probabilistic model used.

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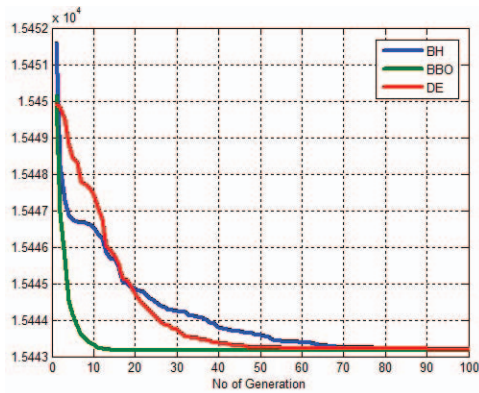


Fig. 5. Mean Convergence Characteristics of minimum cost obtained by using BH, BBO and DE for test system 2 (WPF=0).

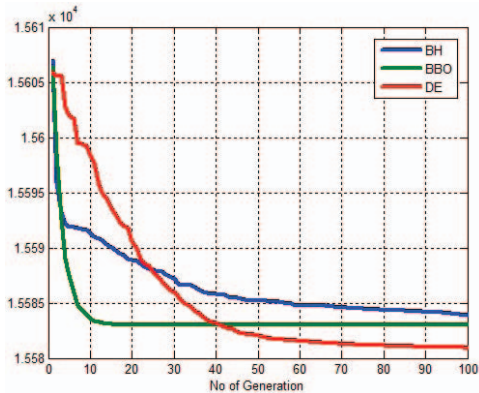


Fig. 6. Mean Convergence Characteristics of minimum cost obtained by using BH, BBO and DE for test system 2 (WPF=50).

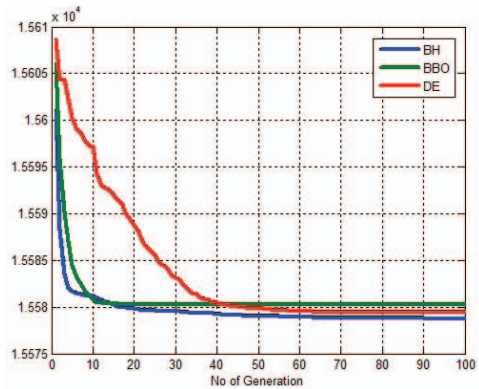


Fig. 7. Mean Convergence Characteristics of minimum cost obtained by using BH, BBO and DE for test system 2 (WPF=165).

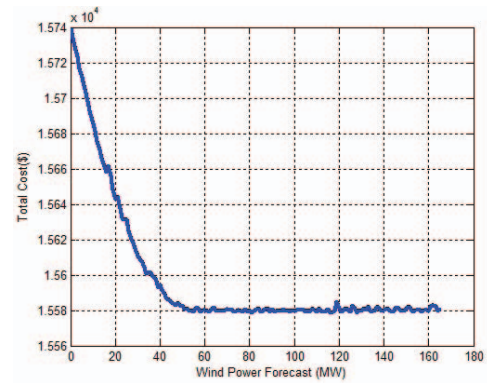


Fig. 8. Total cost variation with respect to wind power generation for test system 2.

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